

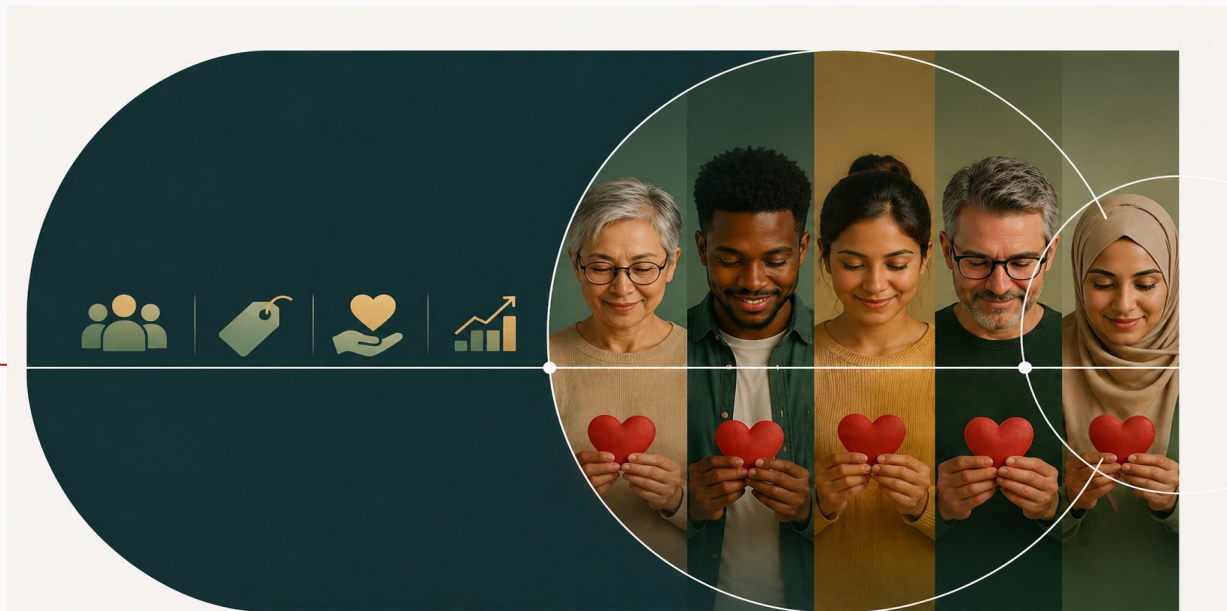
A Comparative Study of Imputation Techniques for Clustering with Incomplete Charity Data

Group 6:

Baixue(Doris) Zhang¹

An Truong²

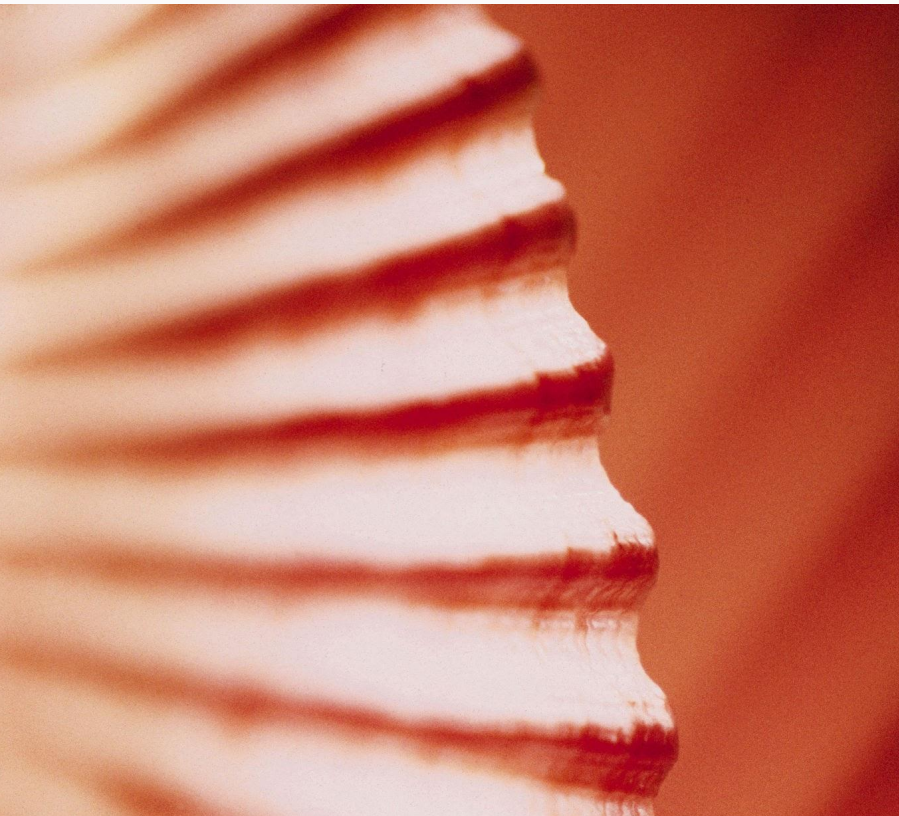
Christie Ngo¹



¹Department of Mathematics and Statistics, Statistics, San José State University

²Department of Computer Science, Data Science, San José State University

Agenda



01 Research Questions & Motivation

02 Data Preprocessing

03 Imputation

04 Baseline Analysis

05 Non Model-Based Partitioning

06 Model-Based Partitioning

07 Model Selection & Conclusion

Research Questions & Motivation

Motivation

- Charitable giving may be related to demographic, political, and religious characteristics.
- Missing survey responses may affect clustering results.

Research Questions

- Does this dataset contain meaningful cluster structure?
- How do different imputation methods affect clustering results?

Approach

- Fill missing values using different imputation methods. $5 + 1 = 6$ datasets.
- Apply different clustering techniques and compare results across datasets.
- Examine whether clusters show interpretable patterns, focusing whether more religious participants tend to donate more.

Data Preprocessing

Dataset Overview

- Online experiment conducted via Amazon Mechanical Turk
- Participants received \$1 with a random multiplier and decided how much to share
- Final dataset contains:
 - ~792 observations
 - 28 variables

Data Cleaning (remove the following variables)

- **Identifier:** AssignmentId, WorkerId, survey code
- **Single-value variables:** Reward, HITTypeid
- **Redundant variable:** e.g., IPAddress, since geographic information is already captured by State.

➡ 14 variables

Clean Variables (14)

- **6 continuous data:** Amount, Multiplier, WorkTime1, WorkTime2, Political Score, Religiosity,
- **5 Categorical data:** Gender, Vote in 2016, State, Party, Batch
- **3 Ordinal data:** Age, income, Religious Attendance

Feature Engineering

- Answers for income such as “Prefer not to state” changed as missing (NA)
- State information aggregated into regions (i.e. NorthWest, West,...)

Missing Data

- Occurs at an observation level
- About 34 % data have 6 missing values (missing follow-up survey)
political views, religious orientation, voting behavior, state, party identity, and religious attendance

Imputation

Imputation

Methods

- **Mean/Mode - 1 imputed dataset**
- **KNN (k = 5, 10)[1] - 2 imputed datasets**
 - K too small may be sensitive to noise
 - K too large may average across dissimilar observations
- **Multivariate Imputation by Chained Equations (MICE) - 2 imputed datasets**
 - PMM (Predictive Mean Matching): for numeric variables (continuous, ordinal)
 - Logistic regression: for binary categorical variables
 - Ordinal logistic regression: for ordered categorical variables
 - Linear regression: for continuous numeric variables
- **One baseline dataset + five imputed datasets.**

Reference:

[1]https://web.stanford.edu/group/herlocklab/pdfs/TROYANSKAYA_BIOINFORMATICS_2001.pdf

Imputation - KNN

Example

Person	Age	Income	Political View	Religiosity
A	25	3	2	?
B	24	3	2	80
C	27	4	2	70
D	30	3	3	60
E	50	8	7	10
F	22	2	1	90

$$d(A, B) = \sqrt{(25 - 24)^2 + (3 - 3)^2 + (2 - 2)^2} = 1$$

$$d(A, C) = \sqrt{5} \approx 2.24$$

$$d(A, D) = \sqrt{26} \approx 5.10$$

$$d(A, E) = \sqrt{675} \approx 25.98$$

$$d(A, F) = \sqrt{11} \approx 3.32$$

Let $k=3$

Nearest neighbor: B, C, F. Religiosity for A is

$$\frac{80 + 70 + 90}{3} = 80$$

Imputation - MICE

Linear Regression Example

Person	Age	Income	Political View	Religiosity
A	25	3	2	?
B	24	3	2	80
C	27	4	2	70
D	30	3	3	60
E	50	8	7	10
F	22	2	1	90

$$\text{Religiosity} = f(\text{Age}, \text{Income}, \text{Political View})$$

Assume:

$$\widehat{\text{Religiosity}} = 110 - 1.2 \times \text{Age} - 2.5 \times \text{Income} + 4 \times \text{Political View}$$

$$A = (25, 3, 2)$$

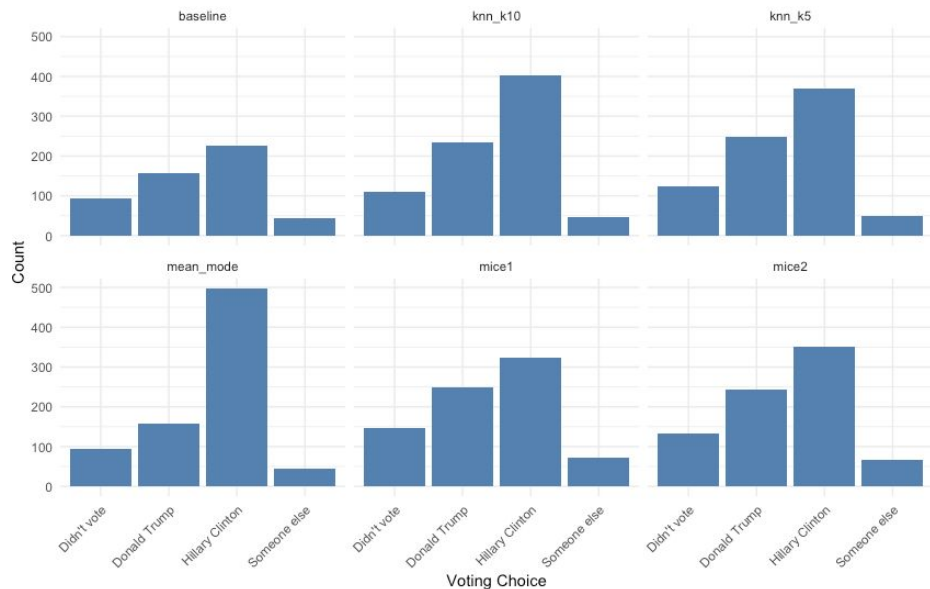
Religiosity for A:

$$\widehat{\text{Religiosity}} = 110 - 1.2(25) - 2.5(3) + 4(2)$$

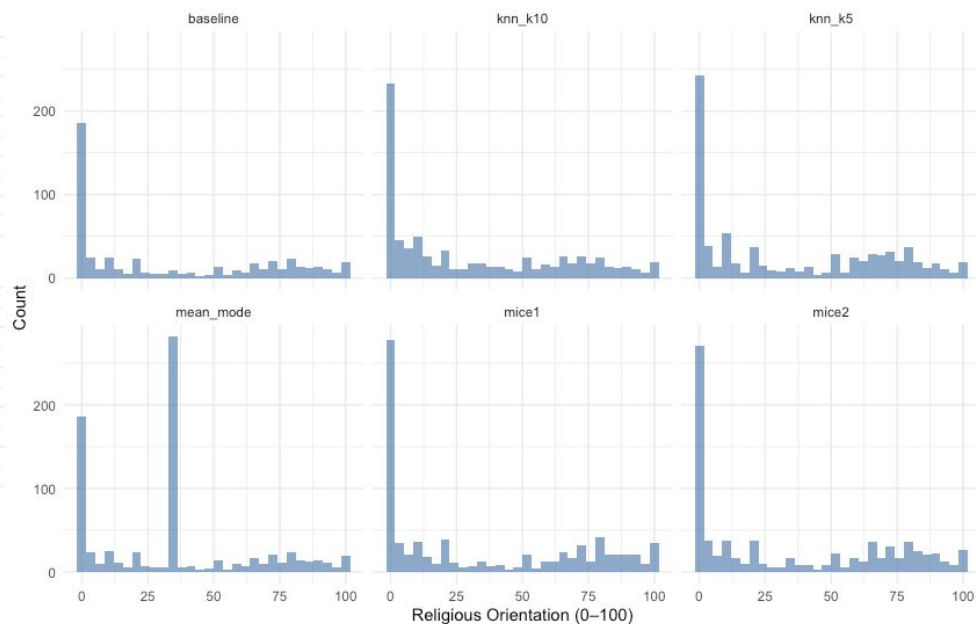
$$= 110 - 30 - 7.5 + 8 = 80.5$$

Imputed Data

Voting Behavior Across Datasets



Distribution of Religious Orientation Across Datasets

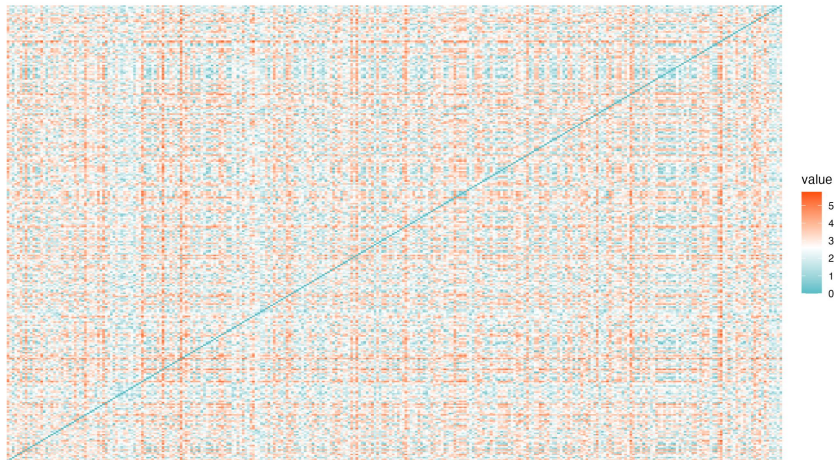


- Overall, the imputed datasets show relatively similar patterns to the baseline.
- Mean/mode imputation noticeably distorts the distributions by concentrating values, while KNN and MICE better preserve the original patterns.

Clustering Methods

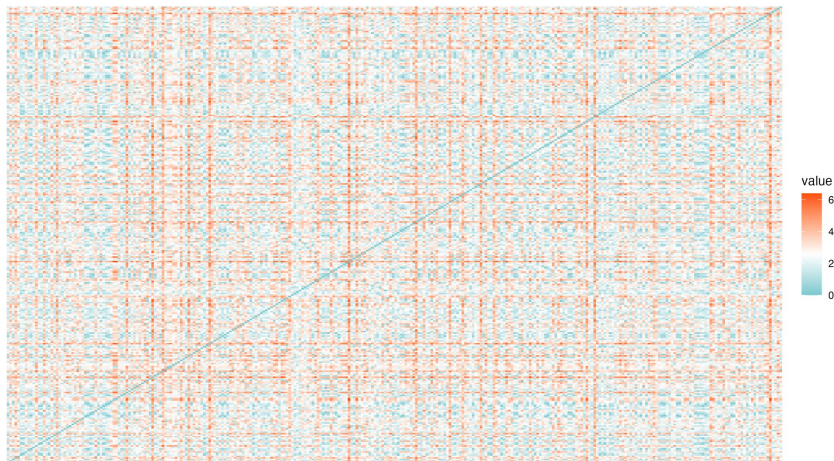
VAT — Continuous | Complete Baseline

Hopkins H = 0.709



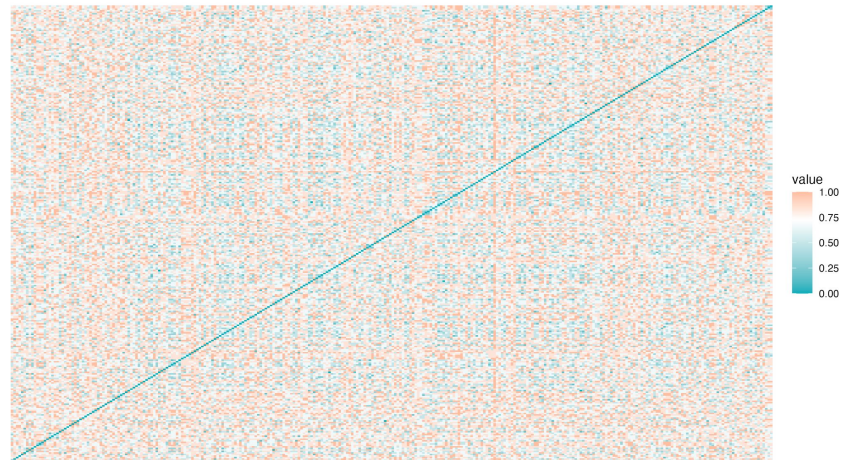
VAT — Continuous | Mean/Mode Imputed

Hopkins H = 0.798



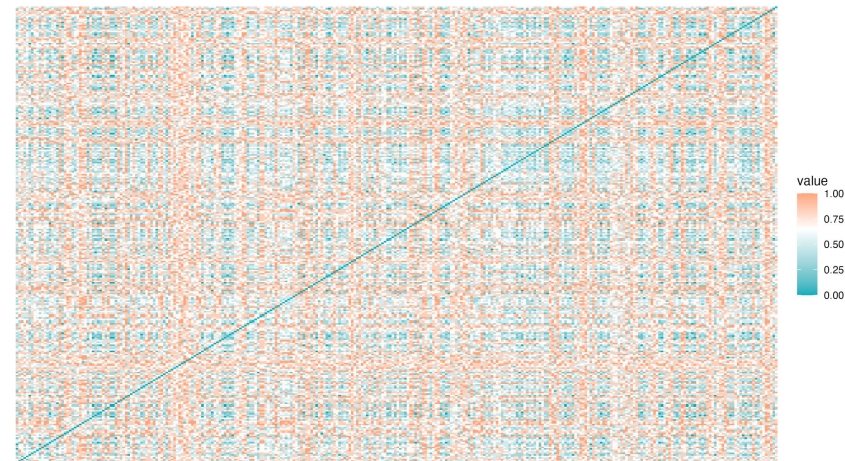
VAT — Categorical | Complete Baseline

Hopkins H = 0.9894



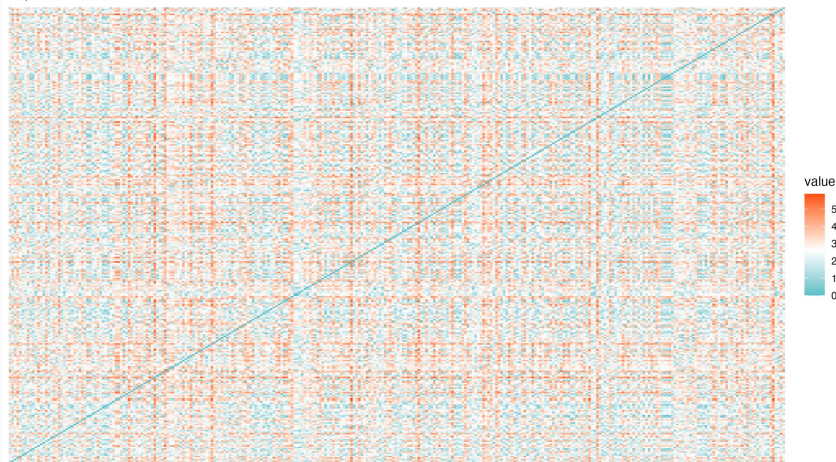
VAT — Categorical | Mean/Mode Imputed

Hopkins H = 0.9907



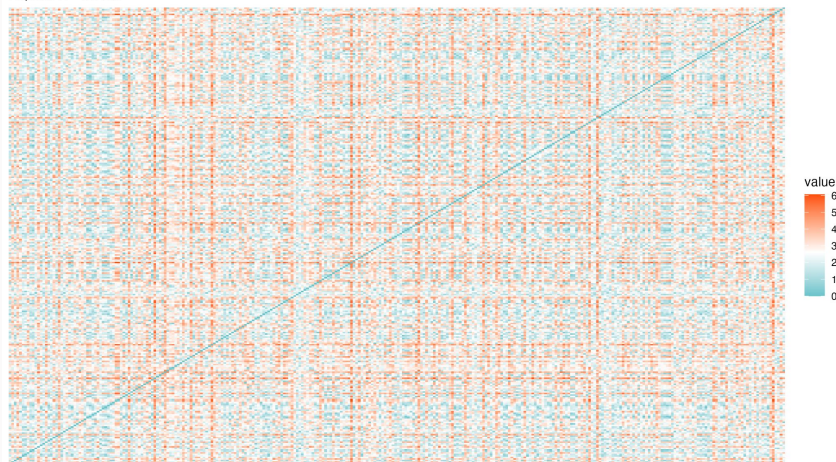
VAT — Continuous I KNN Imputed (k=5)

Hopkins H = 0.7847



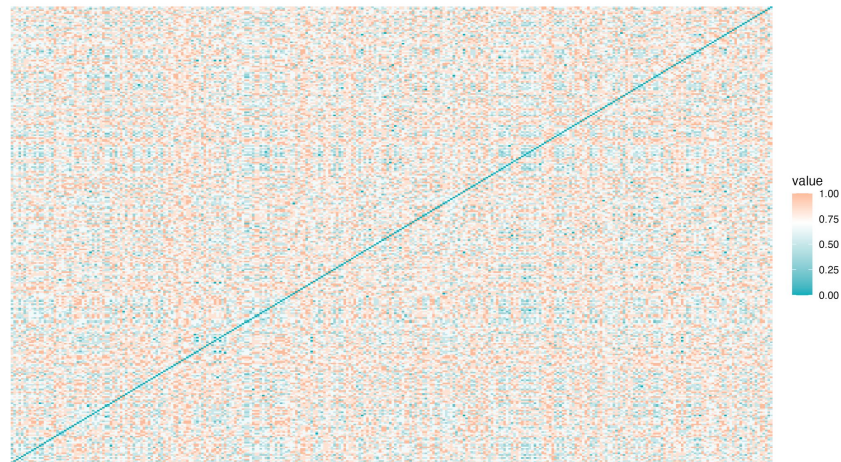
VAT — Continuous I KNN Imputed (k=10)

Hopkins H = 0.7809



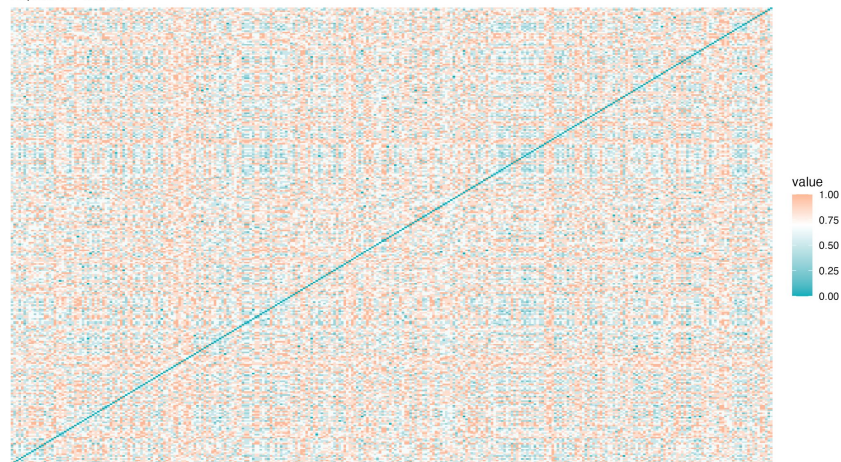
VAT — Categorical I KNN Imputed (k=5)

Hopkins H = 0.9817

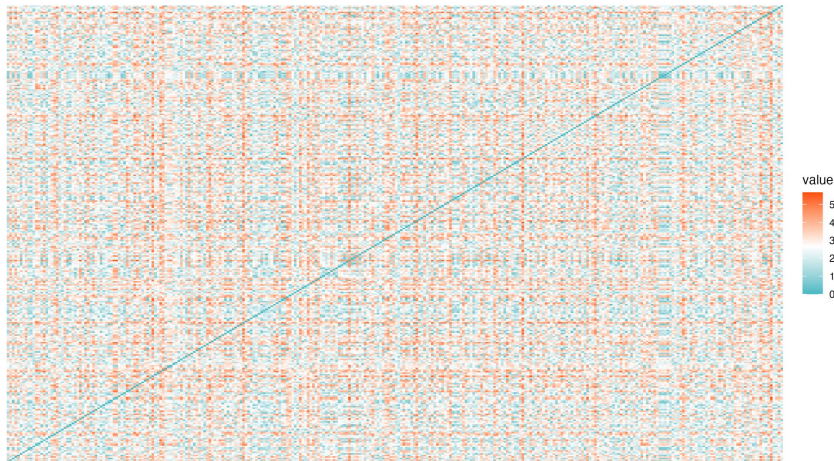


VAT — Categorical I KNN Imputed (k=10)

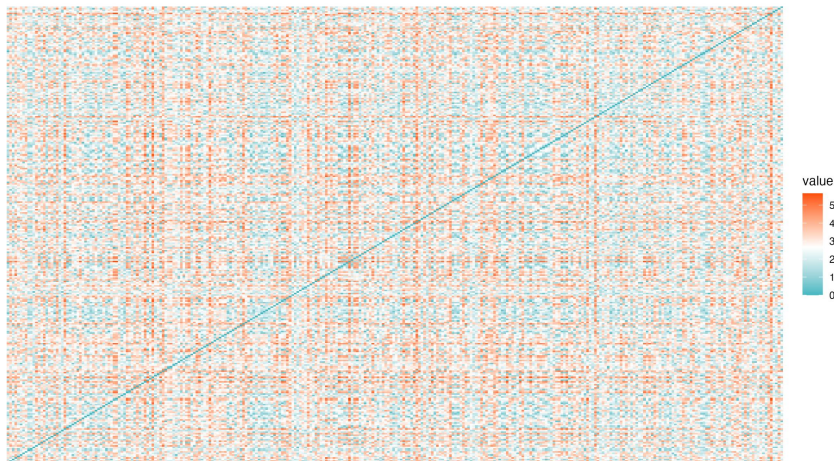
Hopkins H = 0.9848



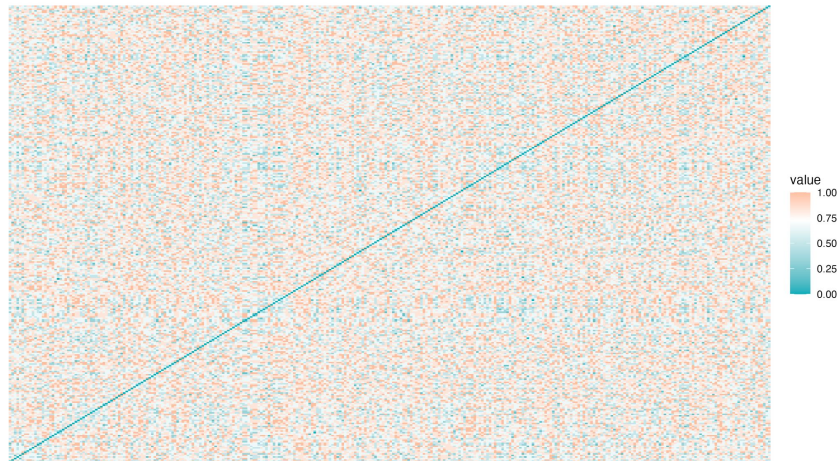
VAT — Continuous I MICE Imputed 1
Hopkins H = 0.7685



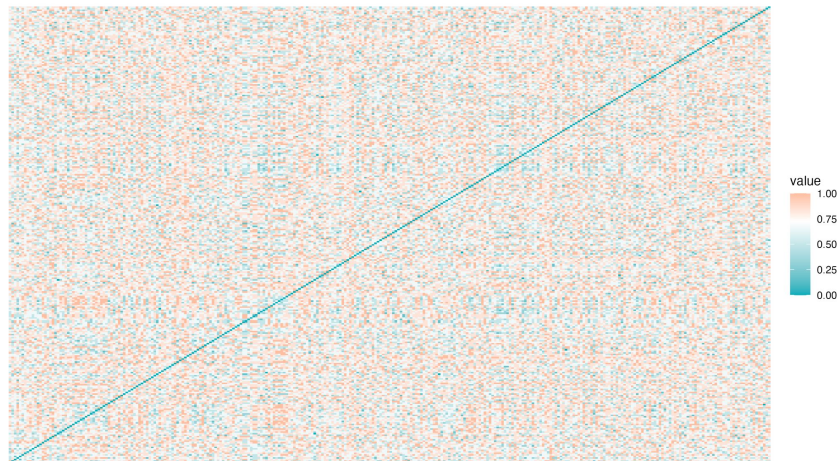
VAT — Continuous I MICE Imputed 2
Hopkins H = 0.7786



VAT — Categorical I MICE Imputed 1
Hopkins H = 0.9847



VAT — Categorical I MICE Imputed 2
Hopkins H = 0.9857



Baseline analysis

Methods

- K-Means
- Fuzzy K-Means (FKM)
- Probabilistic Distance (PD) Clustering
- Hierarchical Clustering
- Generalized Mixture Model (GMM)
- K-Prototype

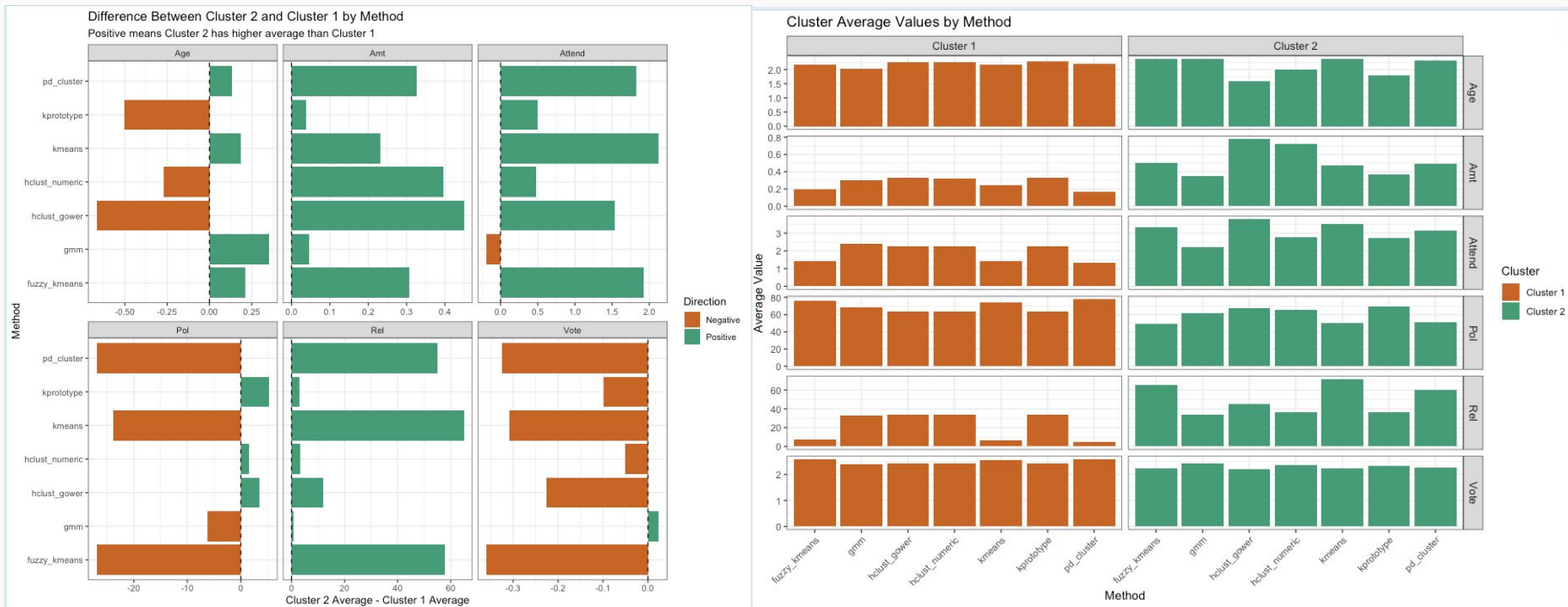
Clustering Methods for Baseline Dataset - ARI

	kmeans	fuzzy_kmeans	pd_cluster	hclust_avg	gmm	kprototype
kmeans	1.0000	0.7847	0.7098	0.0041	0.0007	0.0084
fuzzy_kmeans	0.7847	1.0000	0.8560	0.0014	-0.0003	0.0032
pd_cluster	0.7098	0.8560	1.0000	-0.0002	-0.0006	0.0001
hclust_avg	0.0041	0.0014	-0.0002	1.0000	0.0443	0.3738
gmm	0.0007	-0.0003	-0.0006	0.0443	1.0000	0.1795
kprototype	0.0084	0.0032	0.0001	0.3738	0.1795	1.0000

Takeaways

- Optimal number of cluster K 2, 3, 5 (elbow, silhouette), unstable clustering results.
- For K= 2, there is clean split Indicating strong positive association between religion and giving behavior.

Baseline analysis



Across all datasets, Cluster 2 is generally more religiously oriented and attends services more often, while political and voting differences are less consistent across methods.

Distance-Based Methods

Distance-based (Numerical)

Methods

- K-Means Clustering
- Fuzzy K-Means (FKM)
- Probabilistic Distance (PD) Clustering
- Hierarchical Clustering
 - Linkage: Average

Internal Measures

- Average Silhouette Width (ASW)
- CH Index
- Dunn Index
- Agglomerative Coefficient

Dataset	Method	K	Silhouette	CH_Index	Dunn_Index	Agg_Coef
Complete Baseline	K-Means	8	0.2850	162.2568	0.0294	
Complete Baseline	Fuzzy K-Means	4	0.2613	164.2588	0.0175	
Complete Baseline	PD Clustering	2	0.2453	65.7729	0.0251	
Complete Baseline	Hierarchical Avg	3	0.2593	149.4789	0.0972	0.8815
Mean/Mode Imputed	K-Means	5	0.2882	257.0552	0.0140	
Mean/Mode Imputed	Fuzzy K-Means	4	0.2681	231.0384	0.0112	
Mean/Mode Imputed	PD Clustering	2	0.2072	145.0197	0.0103	
Mean/Mode Imputed	Hierarchical Avg	2	0.3107	170.0892	0.1011	0.9171
MICE Imputed 1	K-Means	7	0.2815	243.5778	0.0206	
MICE Imputed 1	Fuzzy K-Means	2	0.2548	242.0077	0.0442	
MICE Imputed 1	PD Clustering	2	0.2406	119.7539	0.0222	
MICE Imputed 1	Hierarchical Avg	3	0.2473	224.9475	0.0604	0.8969
MICE Imputed 2	K-Means	8	0.2824	243.9648	0.0227	
MICE Imputed 2	Fuzzy K-Means	8	0.2724	210.5918	0.0152	
MICE Imputed 2	PD Clustering	2	0.2390	106.6594	0.0295	
MICE Imputed 2	Hierarchical Avg	3	0.2493	204.6626	0.0933	0.8967
KNN k=5	K-Means	8	0.2906	247.1376	0.0275	
KNN k=5	Fuzzy K-Means	8	0.2789	218.9233	0.0157	
KNN k=5	PD Clustering	2	0.2391	104.5658	0.0115	
KNN k=5	Hierarchical Avg	2	0.2481	147.3748	0.0936	0.8977
KNN k=10	K-Means	6	0.2805	250.8733	0.0193	
KNN k=10	Fuzzy K-Means	4	0.2694	248.4977	0.0166	
KNN k=10	PD Clustering	2	0.2336	121.4763	0.0262	
KNN k=10	Hierarchical Avg	2	0.2660	161.5770	0.0843	0.8978

Distance-based (Numerical)

ARI Comparison Across Imputed Datasets

Comparison	K-means	Fuzzy K-means	PD Clustering	Hierarchical
Mean/Mode vs MICE 1	0.5102	0.1891	0.1221	0.0610
Mean/Mode vs MICE 2	0.4428	0.4013	0.1348	0.3143
Mean/Mode vs KNN k=5	0.4516	0.4088	0.1561	0.6833
Mean/Mode vs KNN k=10	0.7491	0.6671	0.3665	0.8385
MICE 1 vs MICE 2	0.4430	0.1116	0.4957	0.3666
MICE 1 vs KNN k=5	0.4451	0.1037	0.4993	0.0734
MICE 1 vs KNN k=10	0.5443	0.1865	0.4074	0.0457
MICE 2 vs KNN k=5	0.5055	0.4722	0.5174	0.2581
MICE 2 vs KNN k=10	0.4556	0.3602	0.4172	0.2858
KNN k=5 vs KNN k=10	0.5407	0.3805	0.5927	0.6392

Distance-Based (Mixed data)

K-Prototype

- **Internal Measures:** Average Silhouette Width, Elbow method
- **Result**

	knn_k5	knn_k10	mice1	mice2	mean_mode
knn_k5	1.0000	0.9444	0.7144	0.6294	0.9444
knn_k10	0.9444	1.0000	0.6856	0.6397	1.0000
mice1	0.7144	0.6856	1.0000	0.5713	0.6856
mice2	0.6294	0.6397	0.5713	1.0000	0.6397
mean_mode	0.9444	1.0000	0.6856	0.6397	1.0000

Dataset	n	Cluster 1							Cluster 2						
		Amt	Pol	Relig	Age	Inc	Female	Trump	Amt	Pol	Relig	Age	Inc	Female	Trump
KNN k=5	44 / 748	0.367	69.6	39.2	1.73	3.32	0.364	0.409	0.330	63.9	34.0	2.19	3.26	0.477	0.309
KNN k=10	40 / 752	0.386	68.9	37.1	1.78	3.28	0.400	0.373	0.329	64.5	30.6	2.18	3.26	0.475	0.290
MICE 1	69 / 723	0.302	64.9	36.2	1.77	3.29	0.406	0.333	0.335	62.0	34.2	2.20	3.26	0.477	0.313
MICE 2	75 / 717	0.340	68.3	45.7	1.73	3.24	0.400	0.373	0.332	65.4	33.1	2.21	3.26	0.478	0.298
Mean/Mode	40 / 752	0.386	68.8	36.4	1.78	3.25	0.400	0.350	0.329	63.8	33.3	2.18	3.25	0.475	0.190

Distance-Based (Categorical)

Methods

- Multiple Correspondence Analysis (MCA) + k-means algorithm

Internal Measures

- Average Silhouette Width (ASW)

Categorical Dataset	K	ASW
Baseline	4	0.292
MM	4	0.376
Mice 1	4	0.289
Mice 2	4	0.288
KNN-5	4	0.309
KNN-10	4	0.321

Categorical Dataset	MM	Mice 1	Mice 2	KNN-5	KNN-10
MM	1.000	0.232	0.279	0.319	0.410
Mice 1	0.232	1.000	0.471	0.457	0.410
Mice 2	0.279	0.471	1.000	0.481	0.487
KNN-5	0.319	0.457	0.481	1.000	0.708
KNN-10	0.410	0.410	0.487	0.708	1.000

Model-based Partitioning

Methods

- Skewed-t
- Generalized Hyperbolic Distribution (GHD)

Internal Measures

- BIC

Continuous Dataset	Skew t G	Skew t BIC	GHD G	GHD BIC
Baseline	2	6761.286	6	-3377.189
MM	FAILED	N/A	3	294.444
Mice 1	2	9652.748	5	-4511.356
Mice 2	2	10278.560	7	-4744.835
KNN-5	2	10393.910	6	-4606.135
KNN-10	2	9718.209	8	-4035.763

Continuous & Ordinal Dataset	Skew t G	Skew t BIC	GHD G	GHD BIC
Baseline	2	9854.426	7	-7698.030
MM	FAILED	N/A	6	5146.632
Mice 1	2	15026.130	5	-10253.890
Mice 2	2	14934.010	5	-10346.890
KNN-5	2	15017.320	8	-10061.460
KNN-10	3	14451.850	5	-10397.470

Investigation of Clusters

Model Based

- Skewed-t
- Distinct groups along lines of religiosity and political affiliation

Cluster	N	Mean Scores (Continuous)					Proportions (Categorical)				
		Age	Income	Politics	Religion	Time 1	Time 7	Female	Trump	Never Att.	Democrat
Cluster 1	268	1.94	3.14	68.50	32.90	525.0	498.0	0.358	0.369	0.511	0.541
Cluster 2	524	2.28	3.33	62.80	29.90	125.0	60.8	0.529	0.256	0.584	0.542

Cluster	N	Mean Scores (Continuous/Ordinal)						Proportions (Categorical)				
		Age	Amount	Income	Politics	Religion	Time 1	Time 7	Female	Trump	Never Att.	Democrat
Cluster 1	458	2.14	0.359	3.22	57.90	54.20	363.0	450.0	0.459	0.400	0.273	0.362
Cluster 2	334	2.20	0.296	3.31	68.10	7.20	119.0	59.0	0.488	0.198	0.698	0.533

Model Selection & Discussion

Imputation Method Impact

- Imputation choice can meaningfully affect model-based clustering results; KNN and MICE better preserve data structure than mean/mode these methods as well. However, the imputed datasets for k prototypes clustering tend to be stable.

Weak Cluster Signal

- A majority of ARI values were low across method pairs indicating a weak, unstable cluster structure in the data implying clusters may be algorithm-sensitive

Distance-based Methods

- Mean/mode imputation unexpectedly performed best for k-means and FKM by internal validation metrics, likely because KNN/MICE's correlation assumptions don't suit this dataset

Model-Based Methods

- Mean/mode suppresses variation and hinders skewed-t/GHD; KNN is favorable here given the dataset's straightforward demographic structure. MICE is more computationally expensive and would be preferable for larger, more complex datasets

Model Selection & Discussion

Categorical & K-Prototypes

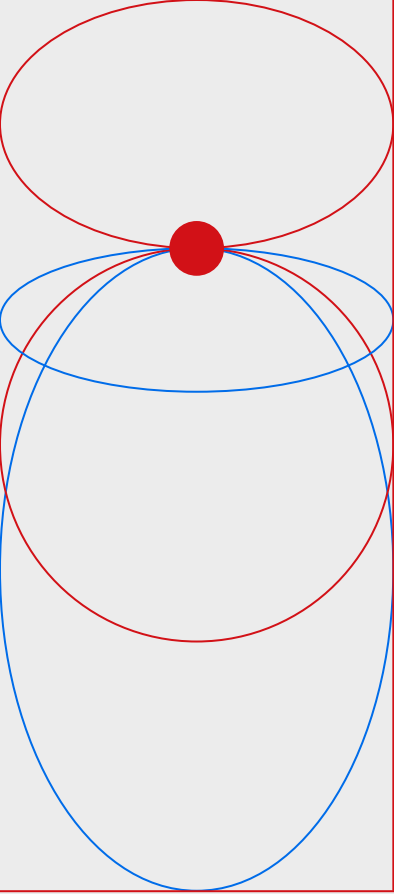
- MCA results were largely stable across imputation methods given the limited category numbers
- K-prototypes showed high cross-imputation consistency, suggesting robust cluster structure for mixed-type methods

Key Takeaways

- Clustering structure was present across all methods, with findings aligning with the author's conclusions (notably religiosity-based donation behavior distinctions).
- Imputation methods can be seen as stable with clustering methods such as K-Prototype but not for others.

Future Work

- Explore MissForest for non-parametric, outlier-robust imputation
- Investigate observation-level stability (core groups that consistently cluster together)
- Expand dataset size to study non-respondent behavior



*Thank
you!*